

Question number	Answer	Additional guidance	Mark
4(a)	An explanation that includes the following points: • allopatric speciation (1) • (allopatric) because lemurs have the same common ancestor isolated from those in Africa (1) • sympatric speciation (1) • (sympatric) because Sifakas and Indri live together but have a different diet (1)	ACCEPT geographic speciation ACCEPT description of how diet is different	(4)

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4(b)(i)	An explanation that includes two of the following points: • because only small quantities of DNA can be collected (1) • therefore need to {amplify / increase the number of copies of} DNA (1) • so that there is enough to {run on gel electrophoresis / analyse} (1)	ACCEPT collecting a sample is difficult	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes two of the following points: • gel electrophoresis run on DNA from both groups of lemurs (1) • pattern of bands were similar showing the lemurs were genetically similar (1) • base sequencing would show similar sequences (1)	ACCEPT use of bioinformatics to show that the base sequences	(2)